Service Design and Innovation
AA 2025 - 2026

WHAT ARE SERVICES?

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01

SERVICES

What are services?

What are services?

The action of helping or doing work for someone.
A **system supplying a public need** such as transport, communications, or utilities such as electricity and water.
Oxford Dictionary

A **government system or private organization** that is responsible for a particular type of activity, or for providing a particular thing that people need (i.e., the postal service).
Cambridge Dictionary

What are services?

“In economic terms, **service occurs when there’s a value exchange between parts**. One part, **the service provider**, performs a certain activity that results in some benefit and includes a specific output that involves certain experiences. The other part, **the service user**, sees value in the output, the experience, or both and is willing to pay for it or exchange for something else of equivalent value”.

Penin, L. (2018). *Designing the Invisible*. London: Bloomsbury.

What are services?

Service is “any direct provision or co-creation of value between a provider and a customer.”

Rust, R.T. and Huang, M. (2014), “The service revolution and the transformation of marketing science”, *Marketing Science, 33*(2), 206-221.

What are services?

1. Services are the soft infrastructure of society and economic exchange.

2. Services have both tangible and intangible elements to them.

1. Services are the soft infrastructure of society and economic exchange.

Service Economy

Services are the soft infrastructure of society and economic exchange.

Emphasis began at the end of the 90s.

“Services are transforming OECD economies on a massive scale (…) With manufacturing slipping to less than 20% of GDP and **the role of services rising to more than 70%** in some OECD countries, services are seen as playing a principal role in economies.”

OECD (2000). *The Service Economy*. Available at: https://www.oecd.org/sti/ind/2090561.pdf

Services are the soft infrastructure of society and economic exchange.

From 2000 onwards services are the largest component of our economy.

Industry has been increasingly transformed by the notion of ***servitization***, with many manufacturing organizations building the service component into their product offerings.

The current list of Fortune 500 companies contains more service companies and fewer manufacturers than in previous decades.

Servitisation

"More and more corporations throughout the world are **adding value to their core corporate offerings through services**. The trend is pervading almost all industries, is customer demand-driven, and perceived by corporations as sharpening their competitive edges. Modern corporations are increasingly offering fuller market packages or **"bundles" of customer-focused combinations of goods, services, support, self-service**, and knowledge. But **services are beginning to dominate. This movement is termed the "servitization of business".**

Vandermerwe, S., Rada, J. (1988). Servitization of business: Adding value by adding services. *European Management Journal*, 6(4), 314–324.

1. Services are the soft infrastructure of society and economic exchange.

What could be examples of companies that have made this shift?

A company that started with products and evolved into services

IBM

Founded in 1911, as the Computing-Tabulating-Recording Company, coinciding with the shift of the United States' agricultural economy to an industrial one.

The company's success at these times was due to the wide range of products it offered that were in high demand in industrializing economies from time recording clocks and commercial scales.

In 1914, the company created the first line of products in the data-processing industry.

Hollerith 1890 tabulating machine with sorting box

IBM

IBM went through the Great Depression and World War II and survived, thanks to its focus on new technology. In 1950, IBM entered the electronic computing market.

In the late half of the past century, the company shifted from manufacturing computers and new technologies to offering IT consulting and integration services. By 2015, the company was generating over 60% of its revenue from services.

Currently focusing on hybrid computing and Artificial Intelligence.

IBM's Deep Blue: the first computer to win a chess match against a human world champion (Garry Kasparov)

Nowadays, an increasing number of companies are born completely service-based.

Name three companies you use weekly that never sold a product to you.

These companies were born service-based. They design access, coordination, and behaviour — not products.

- No ownership
- No physical product
- Continuous data exchange
- Behavioural design

Servitisation

"As businesses began **offering solutions instead of products**, it became evident that selling spare parts and after-sales services—conducting repairs; installing upgrades; reconditioning equipment; carrying out inspections and day-to-day maintenance; offering technical support, consulting, and training; and arranging finances—could be a bountiful source of revenues and profits as well.
How bountiful? In industries such as automobiles, white goods, industrial machinery, and information technology, companies have sold so many units over the years that **their aftermarkets have become four to five times larger than the original equipment businesses**."

Cohen, M. A., & Agrawal, N. (2006). Winning in the Aftermarket. *Harvard Business Review*, 84, 129–138.

Servitisation

It’s the shift from product manufacturers to solutions providers.

Product

- Tools can only be owned
- User as **receiver** – users’ competence applied only to product fruition
- The final activity (product use) is carried out mainly by users

Service

- Tools can be used, without ownership
- User as **co-creator** – creation is shared between user and supplier
- The final activity is divided between user and supplier

Digital Servitisation

Digital servitisation uses digital service technologies to create customer value (Sklyar et al., 2019).

Digital servitization is transforming processes, capabilities and offerings to capture service value arising from a broad range of enabling technologies (Sjödin et al., 2020).

Firms successful in their digital servitization efforts cocreate value with customers through market efficiencies and **data assimilation** (Huang and Rust, 2018).

Digital Servitisation

Holmström and Partanen (2014) define digital servitization in terms of the provision of **digital to act as a substitute for physical goods**.

Kowalkowski et al. (2017, p. 8) frame digital servitization as the utilization of digital tools for “the transformational processes whereby a company shifts from a product-centric to a service-centric business model and logic.”

Sjödin et al. (2020, p. 479) define digital servitization as the “**transformation in processes, capabilities, and offerings within industrial firms and their associate ecosystems to progressively create, deliver, and capture service value arising from a broad range of enabling digital technologies.**”

Nowadays, an increasing amount of companies are born completely service-based.

Additionally, these companies are also born digital first.

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A company is born digital-first when:

- **Its core operations are algorithmic.**
- **Data is the main resource.**
- **Interaction happens primarily through digital interfaces.**
- **Scaling depends on software, not physical expansion.**
- **Without digital infrastructure, the service cannot function.**

Why is Spotify a good example of this type of companies?

Why is Spotify a good example of this type of companies?

Spotify is not digital marketing on top of music;
it is software organising access to music.
It is not a “music company using digital;”
it is a data-driven recommendation system delivering audio.

If Spotify, Uber, Revolut and Airbnb don’t sell products — what are they actually exchanging?

If Spotify, Uber, Revolut and Airbnb don’t sell products — what are they actually exchanging?

They obviously don’t exchange objects.
They exchange the application of competences for the benefit of others.

If Spotify, Uber, Revolut and Airbnb don’t sell products — what are they actually exchanging?

Even when companies do sell objects — like Apple — what people value is not the object itself, but what it enables. We buy communication, coordination, entertainment, identity.
So maybe the object was never the real unit of value.

2. Services have both tangible and intangible elements to them

Service-Dominant Logic:
Helps us understand that the object was never the real unit of value.
Service-Dominant Logic suggests that service — not goods — is the fundamental basis of exchange.

Service-dominant logic

Service-dominant (S-D) logic, in behavioral economics, is an **alternative theoretical framework for explaining value creation**. The underlying idea of S-D logic is that humans apply their competences to benefit others and reciprocally benefit from others' applied competences through service-for-service exchange.

Focusing on service brings attention to the process, patterns, and benefits of exchange, rather than to the units of output that are exchanged (e.g., goods).

Service-dominant logic

Value creation occurs in networks in which resources are exchanged among multiple actors and is therefore more accurately conceptualized as **value co-creation** (Vargo and Lusch, 2008, Vargo, Maglio and Akaka, 2009).

Vargo and Lusch (2004) also distinguish between **service in the singular**, as a fundamental activity of economic exchange, and **services in the plural**, as an economic category in contrast to goods.

Service-dominant logic

	S-D logic axioms and foundational premises
Axiom 1/FP1	Service is the fundamental basis of exchange.
FP2	Indirect exchange masks the fundamental basis of exchange.
FP3	Goods are a distribution mechanism for service provision.
FP4	Operant resources are the fundamental source of strategic benefit.
FP5	All economies are service economies.
Axiom 2/FP6	Value is cocreated by multiple actors, always including the beneficiary.
FP7	Actors cannot deliver value but can participate in the creation and offering of value propositions.
FP8	A service-centered view is inherently customer oriented and relational.
Axiom 3/FP9	All social and economic actors are resource integrators.
Axiom 4/FP10	Value is always uniquely and phenomenologically determined by the beneficiary.
Axiom 5/FP11	Value cocreation is coordinated through actor-generated institutions and institutional arrangements.

Service-dominant logic

Key takeaways of S-D logic:

- it is always fundamentally service, rather than goods, that actors exchange
- value is not created only top-down (i.e., firms are not the sole creator of value)
- value is always co-created through the interaction of actors, either directly or indirectly in service systems

Service-dominant logic + digital servitisation

The rise of digital servitization implies that service exchange and experience extends **beyond human-to-human interactions** (Kleinaltenkamp et al., 2012; Caridà et al., 2019) and may involve the consumer interacting with AI-powered actors (i.e. virtual assistants, chatbots).

Even in predominantly human-to-human service exchanges, the service provider may integrate AI-enabled systems to enhance the service process and thus impact the consumer experience.

Service-dominant logic + digital servitisation

Axiom	Axiom description	Research implications
Axiom 1/FP1	Service is the fundamental basis of exchange	How does the service experience evolve as AI is integrated?
Axiom 2/FP6	Value is co-created by multiple actors, always including the beneficiary	How do service providers increase consumer participation and usage intent of AI-enabled service technologies?
Axiom 3/FP9	All social and economic actors are resource integrators	Who are the actors involved? Is AI an actor? What are the interdependent roles of these actors? How do these roles affect the resource integration process of the value chain?
Axiom 4/FP10	Value is always uniquely and phenomenologically determined by the beneficiary	What is value in AI-enabled service ecosystems? How is value measured? What other perspectives exist for measuring AI-enabled value?
Axiom 5/FP 11	Value cocreation is coordinated through actor-generated institutions and institutional arrangements	What norms or underlying mechanisms exist? How do these facilitate or hinder value co-creation? How does this differ based on the service context, AI interaction level, etc.?

Note: Axiom definitions as stated by Vargo and Lusch (2016, p. 18)

Table 1. Research implications of SD logic axioms for AI-enabled service ecosystems

Service-dominant logic + digital servitisation

Manser Payne, Dahl and Peltier (2020) update the 5 axioms of SD-Logic in light of digital servitisation.

- Axiom 1 contends that AI may offer specialized knowledge that will transform the fundamental nature of service exchange.
- Axiom 2 suggests that consumers need to be active participants to maximize value co-creation.
- Axiom 3 implies that resource integration from a variety of ecosystem of actors needs to be considered.
- Axiom 4 requires that “value” outcomes be measured from the beneficiary’s perspective.
- Axiom 5 outlines that research must consider the ecosystem’s institutions and institutional arrangements that may facilitate or hinder value cocreation.

Service-dominant logic + digital servitisation

Axiom 1 contends that **AI may offer specialized knowledge that will transform the fundamental nature of service exchange**.

Examples:
AI-Powered Virtual Assistants
AI-Powered Healthcare Diagnostics
AI-Powered Financial Advisory Services
AI-Powered E-commerce Recommendations
…

Tension: AI-powered services govern behaviour.

Migliore 28 min 34 min 1h 19m 26 min

Lavoro (Via Giuseppe Candiani, 72)

Duomo M1 M3, 20123 Milano MI

Partenza adesso

Opzioni

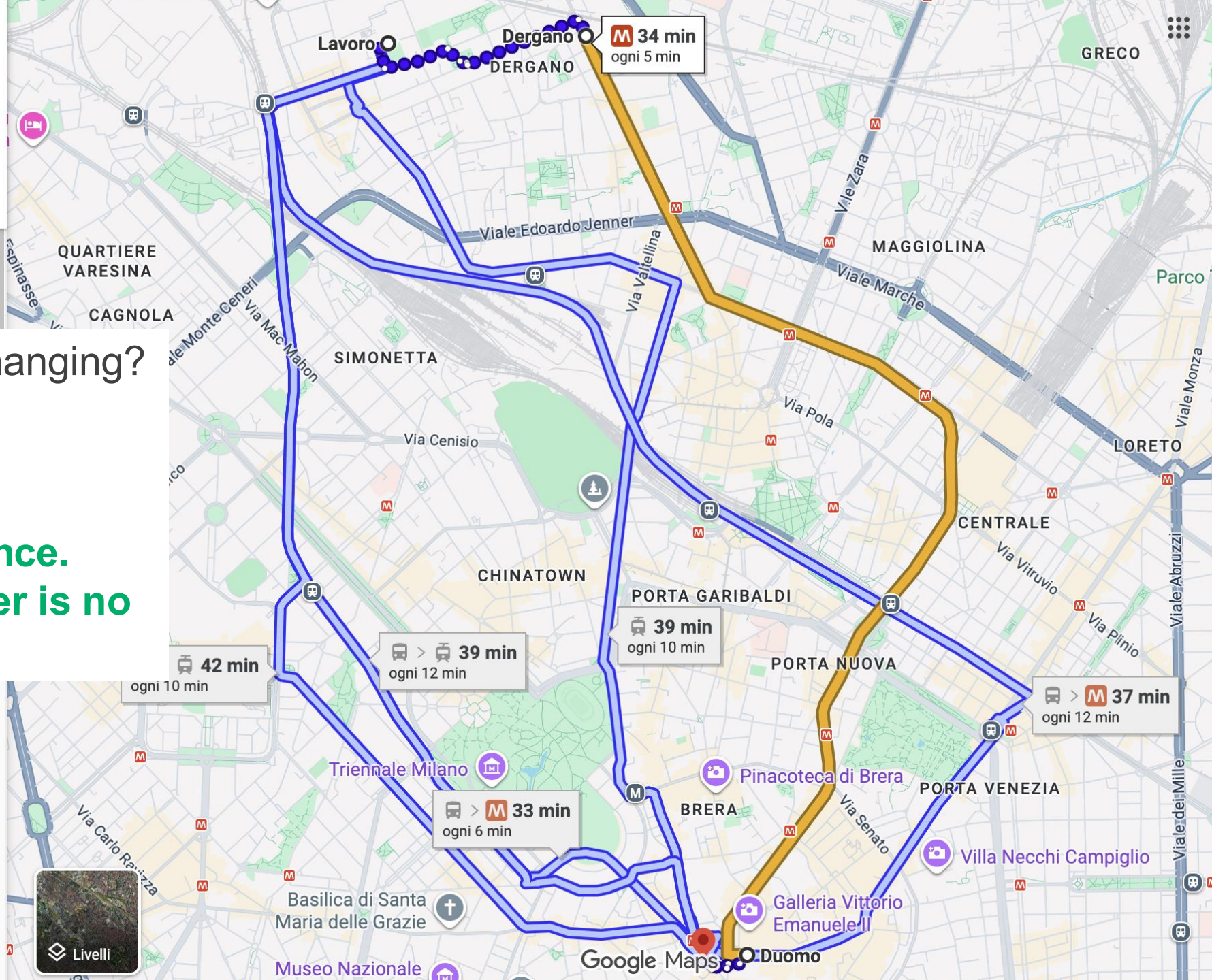
What is Google Maps exchanging?

Not a product.
Not maps.
It's navigational intelligence.
The "knowledge" provider is no longer human.

18:58-19:31 33 min

R16 R17 R22 S3 RE 54 M 1

Novità! Continua il tuo viaggio e tocca la notifica sullo smartphone per ricevere indicazioni stradali



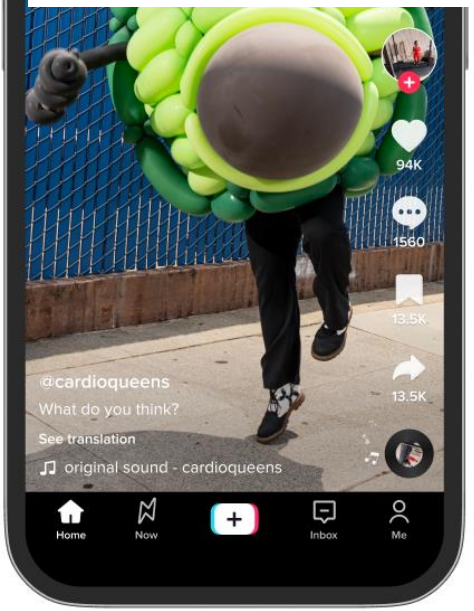
Service-dominant logic + digital servitisation

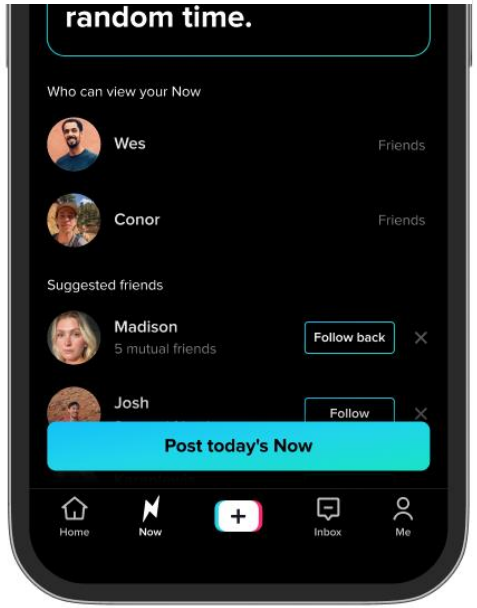
Axiom 2 suggests that **consumers need to be active participants to maximize value co-creation**.

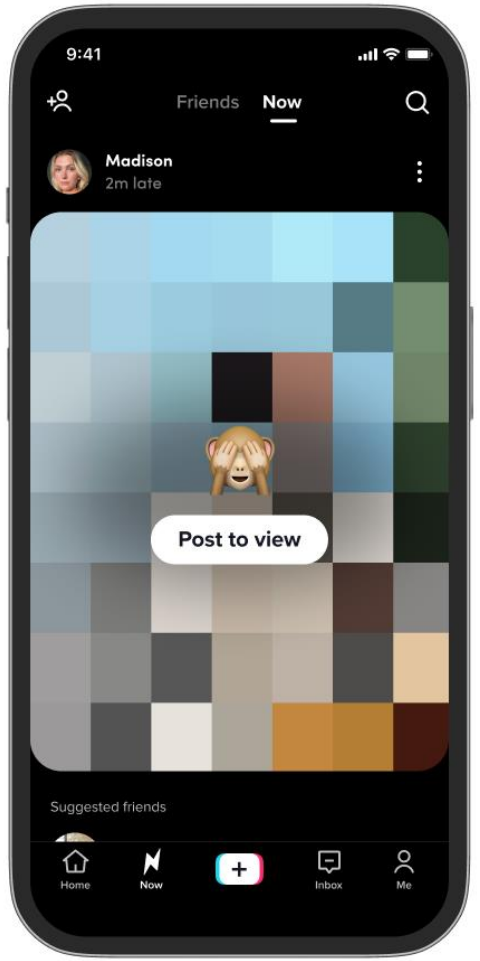
This means that value is co-created by multiple actors, always including the beneficiary.

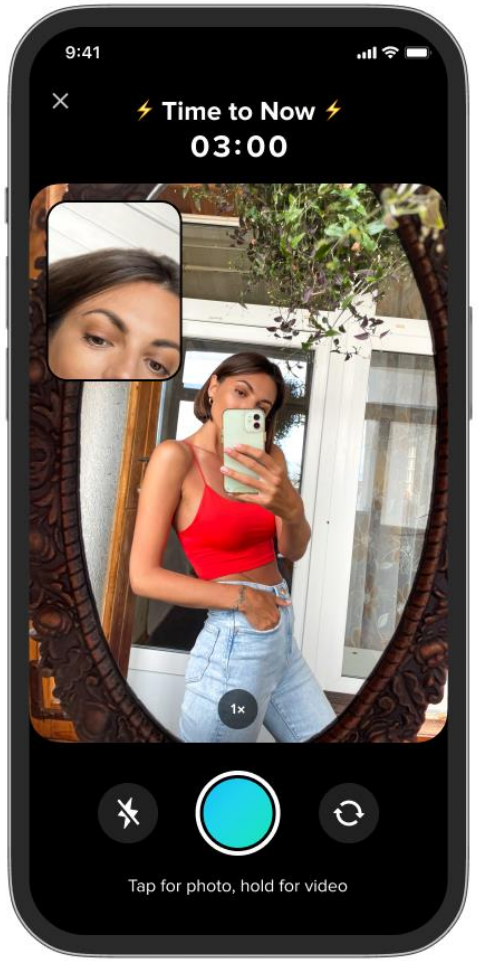
On TikTok, **you are not just a consumer.**
You are raw material for value creation.

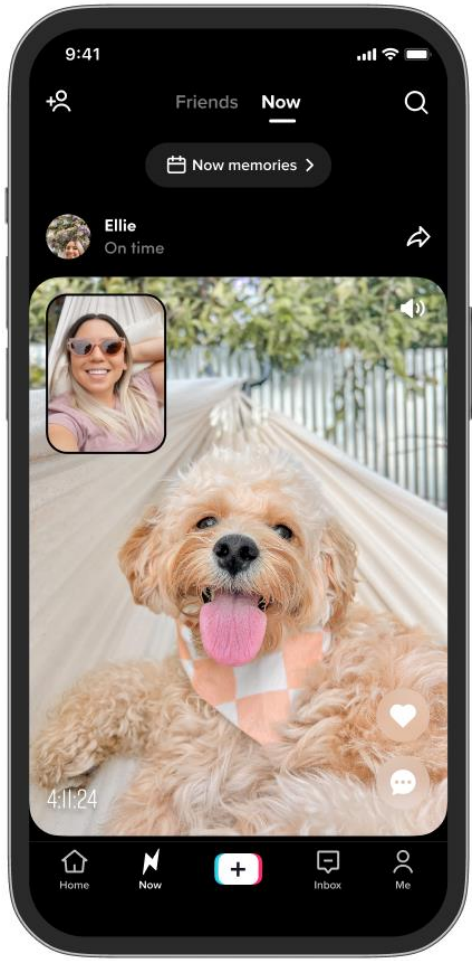
Without your active/constant participation, such a business ideas would never work.









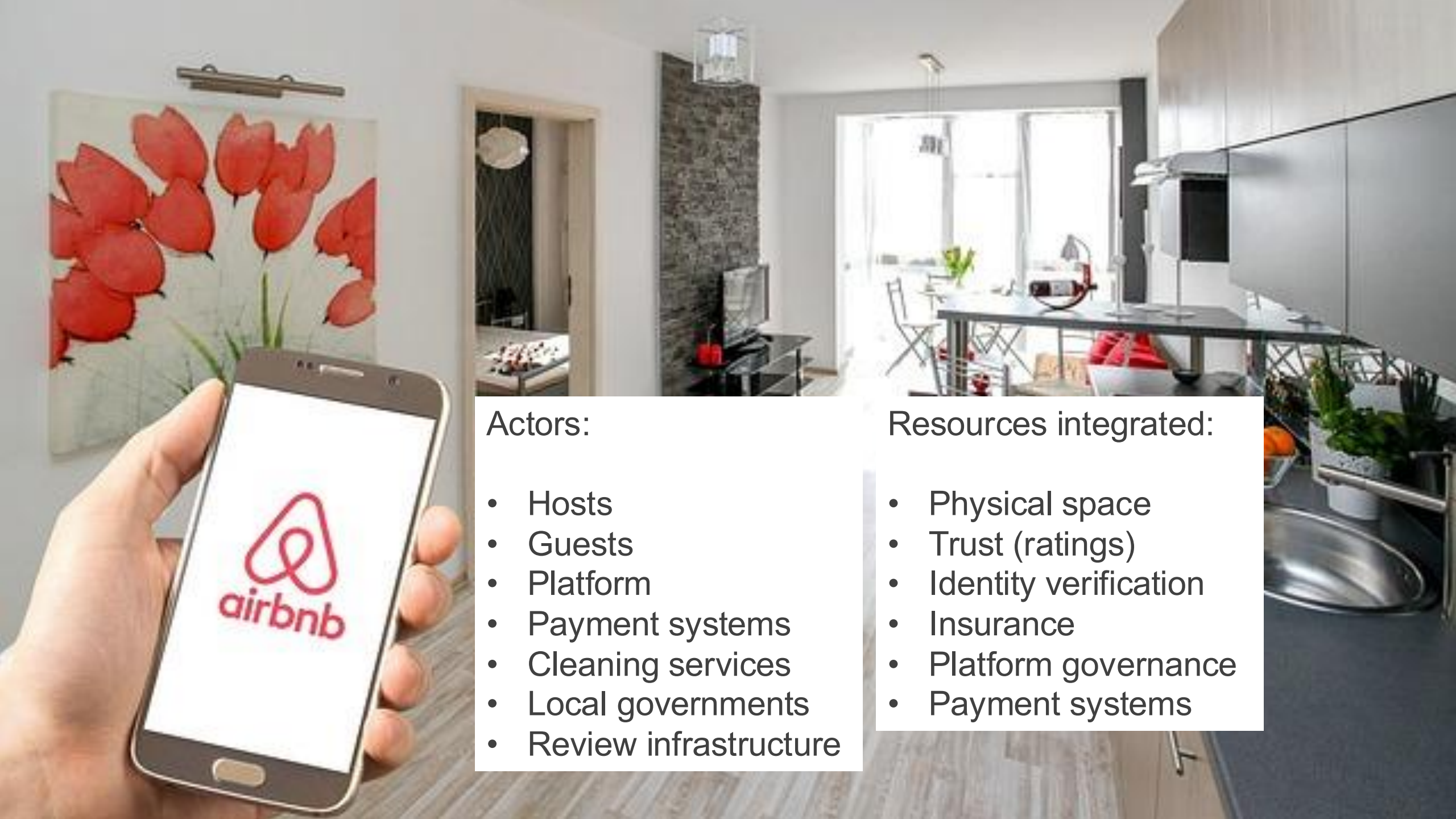


Service-dominant logic + digital servitisation

Axiom 3 implies that **resource integration from a variety of ecosystem of actors needs to be considered**.

This means that no service can be created and offered by one organization only; all actors involved integrate resources from many networks and ecosystems.

Tension: the digital redistributes agency but also agency is no longer only human.

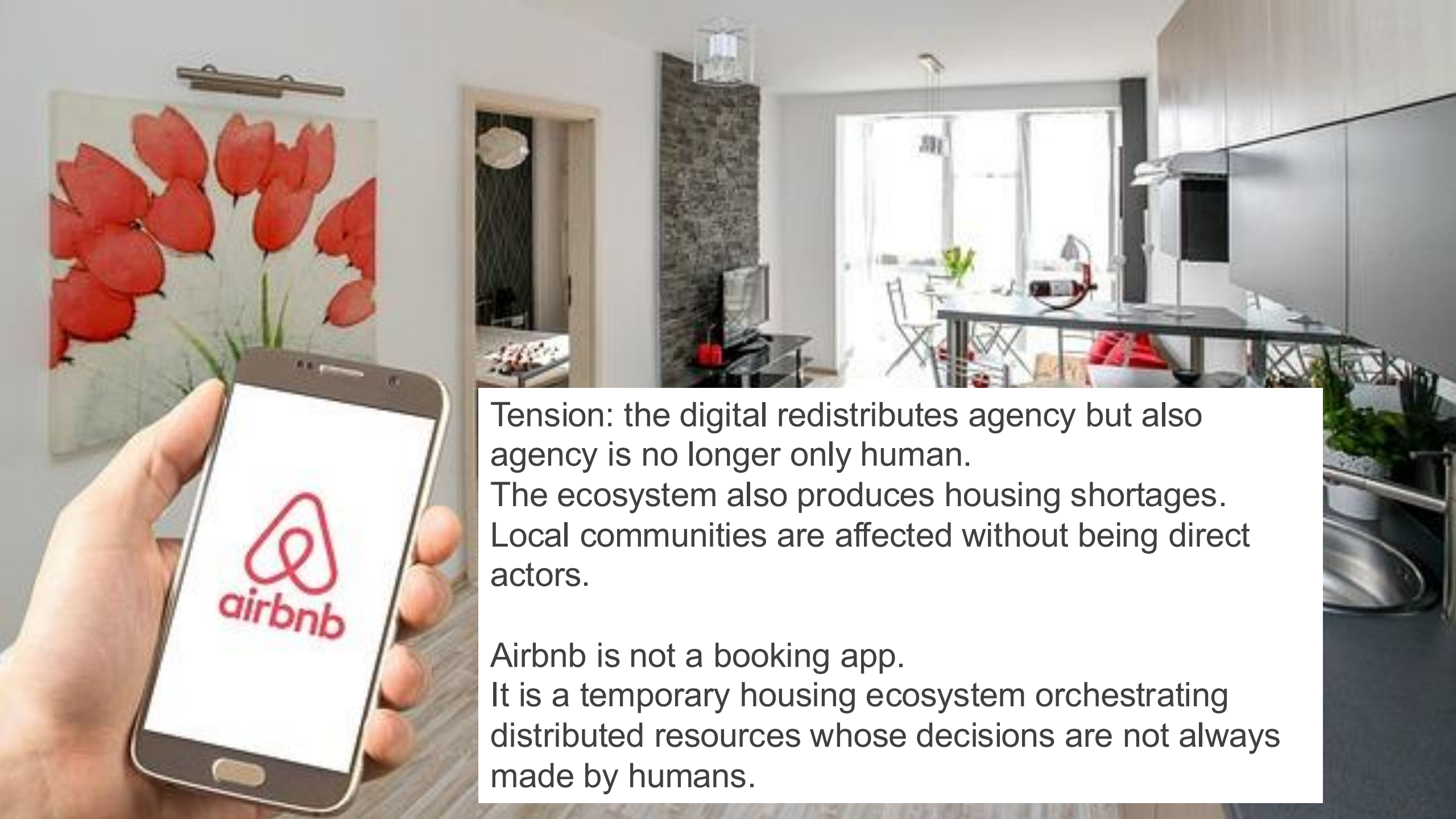


Actors:

- Hosts
- Guests
- Platform
- Payment systems
- Cleaning services
- Local governments
- Review infrastructure

Resources integrated:

- Physical space
- Trust (ratings)
- Identity verification
- Insurance
- Platform governance
- Payment systems



Tension: the digital redistributes agency but also agency is no longer only human.
The ecosystem also produces housing shortages.
Local communities are affected without being direct actors.

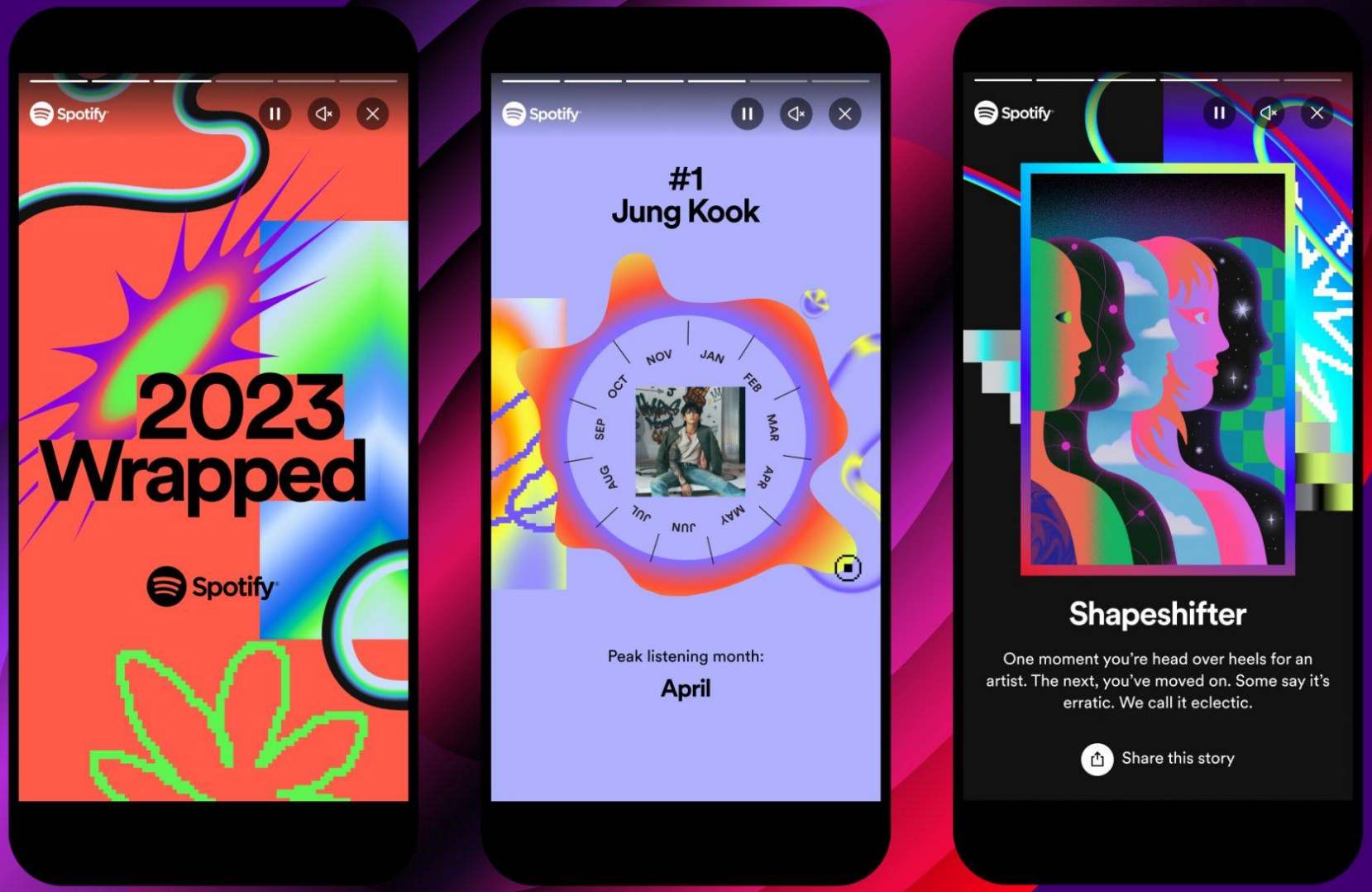
Airbnb is not a booking app.
It is a temporary housing ecosystem orchestrating distributed resources whose decisions are not always made by humans.

Service-dominant logic + digital servitisation

Axiom 4 requires that **“value” outcomes need to be measured from the beneficiary’s perspective**.

This means that value must be evaluated from the user’s perspective — but digital systems measure it.

Tension: AI segments users.


Spotify
2023
Wrapped
Spotify
Spotify
#1
Jung Kook
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
Peak listening month:
April
Spotify
Shapeshifter
One moment you're head over heels for an artist. The next, you've moved on. Some say it's erratic. We call it eclectic.
Share this story

Value for user:

- Entertainment
- Identity expression
- Social sharing
- Personal nostalgia

Value for Spotify:

- Engagement metrics
- Social media amplification
- Data enrichment

Tension:
The beneficiary defines value emotionally.
But the platform measures value numerically.
Spotify Wrapped feels personal.
But it is also a behavioural analytics report.

Service-dominant logic + digital servitisation

Axiom 5 outlines that **services must consider the ecosystem of institutions and institutional arrangements** that may facilitate or hinder value cocreation.

This means that value co-creation is coordinated through institutions and institutional arrangements.

Tension: **platforms + AI** create new asymmetries.

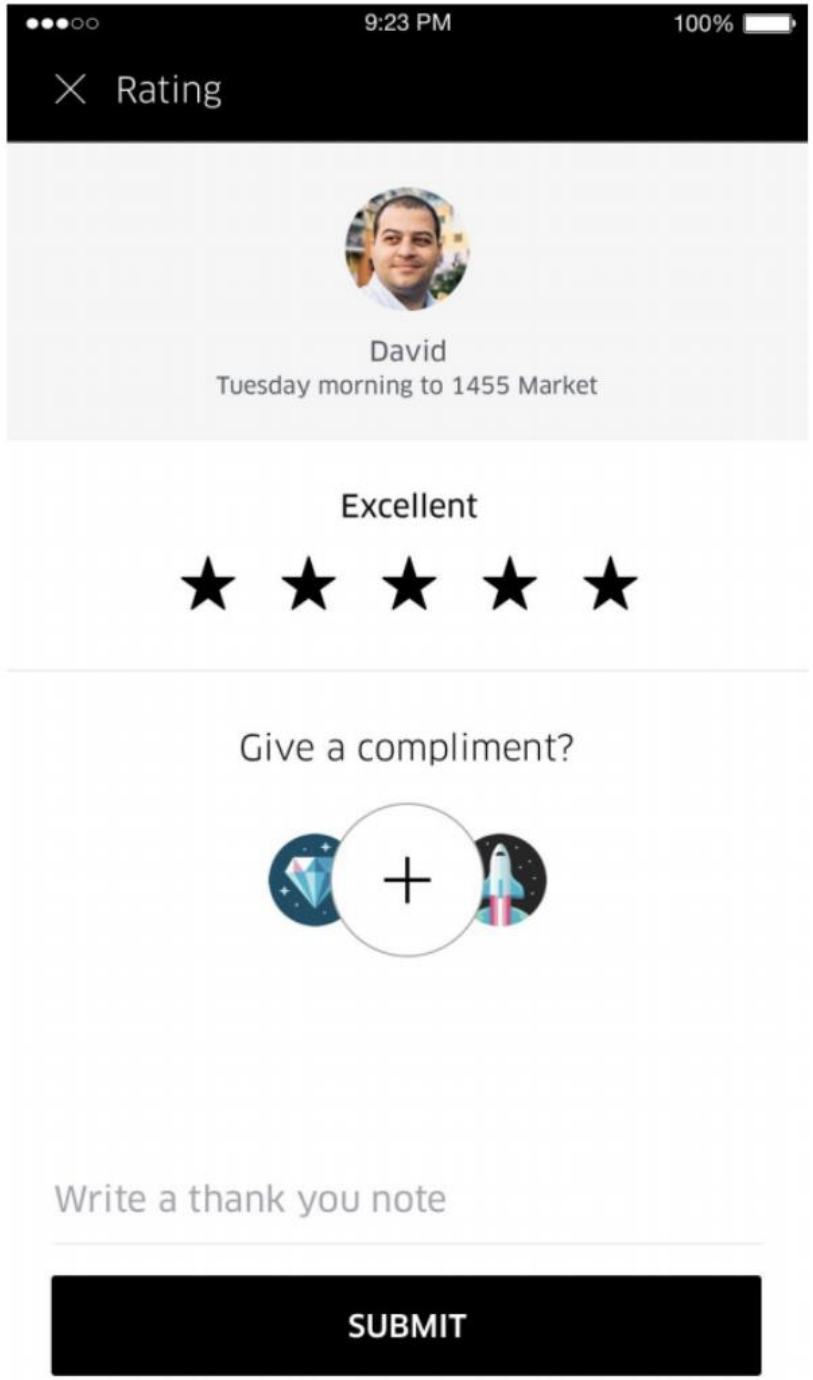
Institutional rules embedded in code:

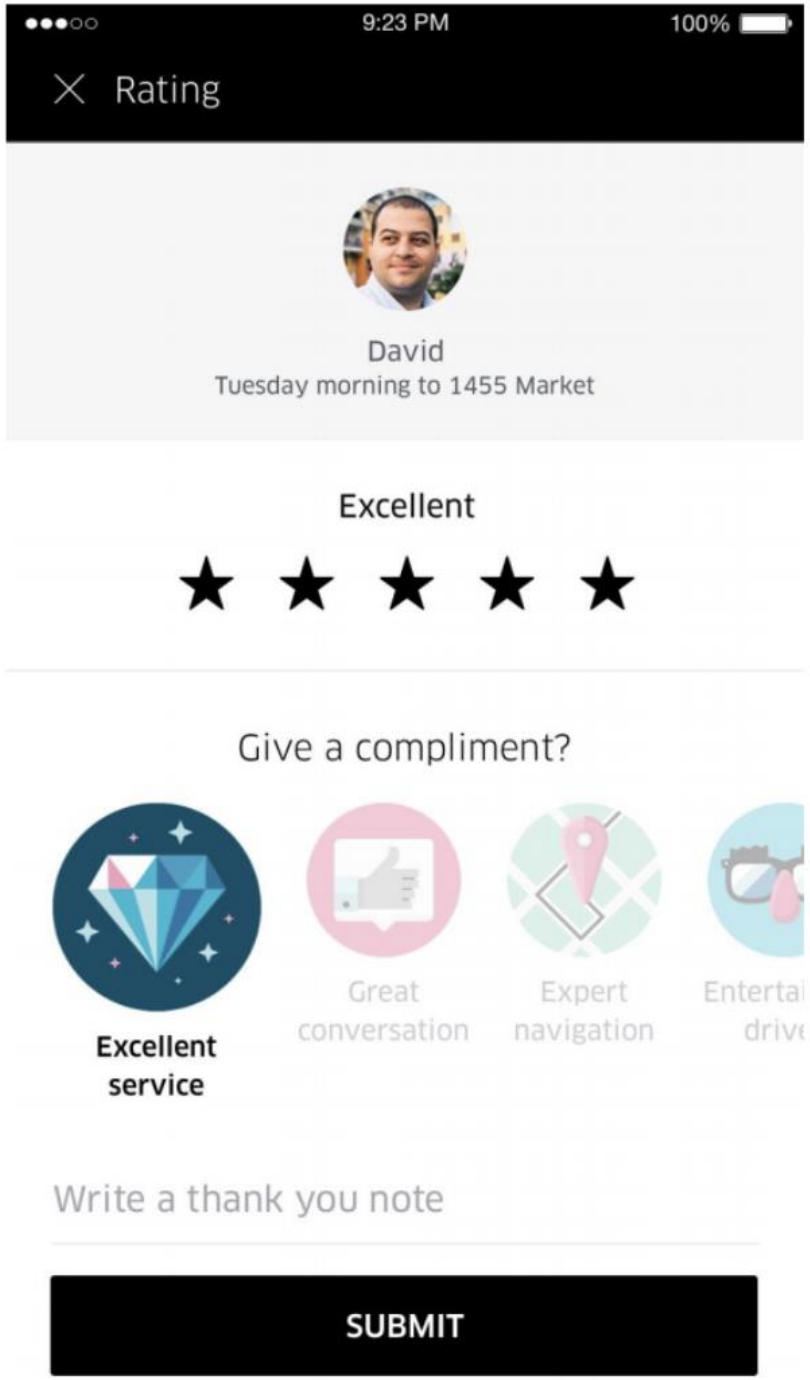
- Drivers must maintain rating above X.
- Riders can be deactivated.
- Surge pricing logic.
- Cancellation penalties.

These are not just features.
They are:

- Governance mechanisms.
- Behavioural regulators.
- Digital labour law.

Uber's rating system is not feedback.
It is institutionalised behavioural governance.





02

SERVICES + DESIGN

If everything is service, what do designers design?

Design

A few definitions – **WHO ARE THE AUTHORS?**

To design is to devise courses of action aimed at changing existing situations into preferred ones.

Design is a reflective practice. In “reflection-in-action”, doing and thinking are complementary. Doing extends thinking in the tests, moves, and probes of experimental action, and reflection feeds on doing and its results. Each feeds the other, and each sets boundaries for the other.

Design is a liberal art capable of dealing with “wicked problems” for which there is no single solution and in which stakeholders play roles in defining the nature of problems.

Design

A few definitions – **WHO ARE THE AUTHORS?**

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Design is a reflective practice. In “reflection-in-action”, doing and thinking are complementary. Doing extends thinking in the tests, moves, and probes of experimental action, and reflection feeds on doing and its results. Each feeds the other, and each sets boundaries for the other. (Schön, 1987)

Design is a liberal art capable of dealing with “wicked problems” for which there is no single solution and in which stakeholders play roles in defining the nature of problems. (Buchanan, 1992)

Design

(Simon, 1969) (Schön, 1987)

Deterministic

Design as problem solving.

The aim is to work towards a desired state of affairs that can be determined in advance.

Exploratory enquiry

Design is a situated activity, influenced by context and culture in which action takes place.

Knowledge and solutions emerge while working on an issue.

Design

Evolution in scope and object

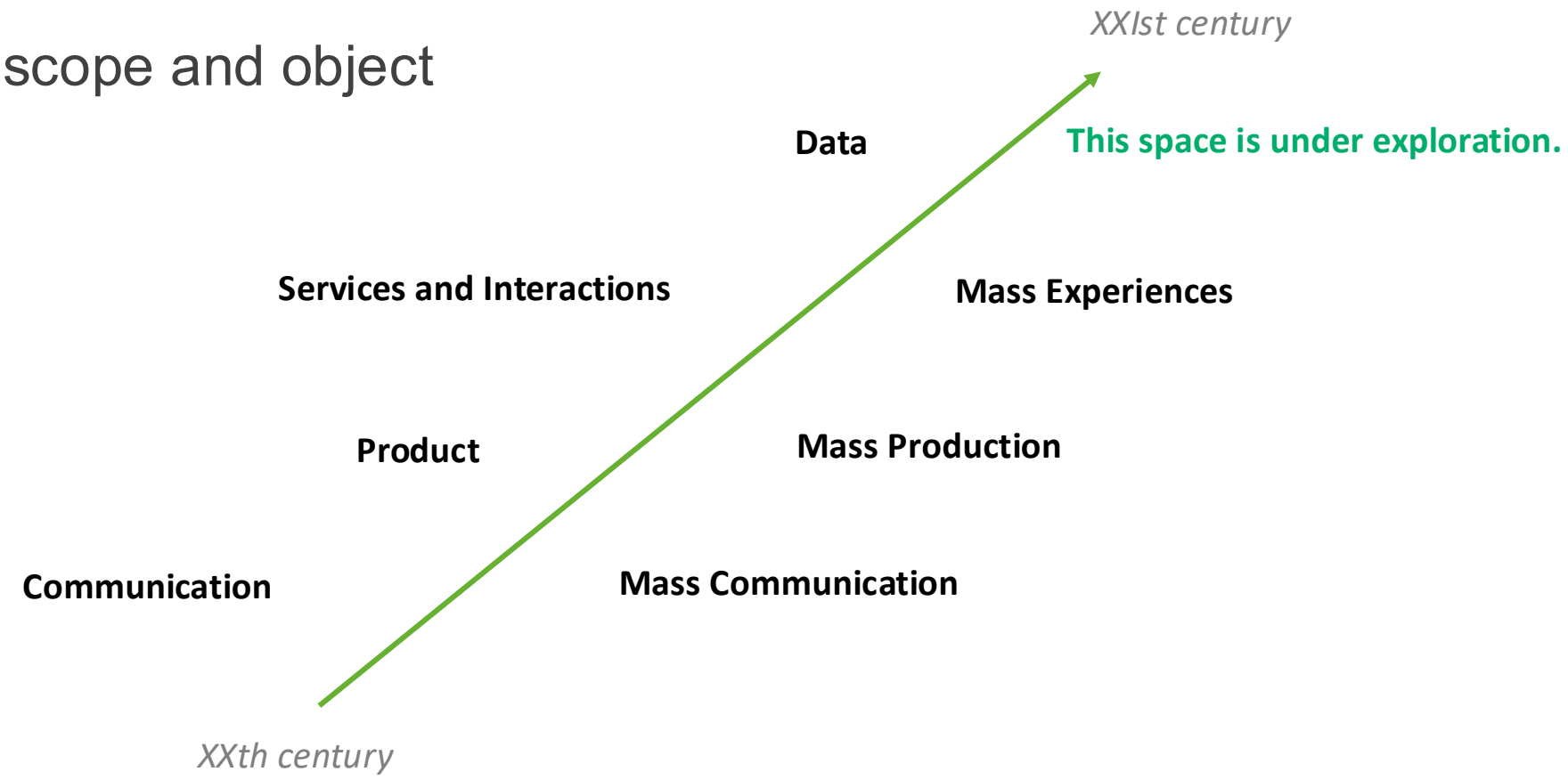
Fields of Design Problems

Arts of Design Thinking	Communication Symbols	Construction Things	Interaction Action	Integration Thought
Inventing Symbols	Symbols: Words & Images			
Judging Things		Physical Objects		
Connecting Action			Activities, Services, Processes	
Integrating Thought				Systems, Organizations, Environments

(Source: Buchanan, 2001)

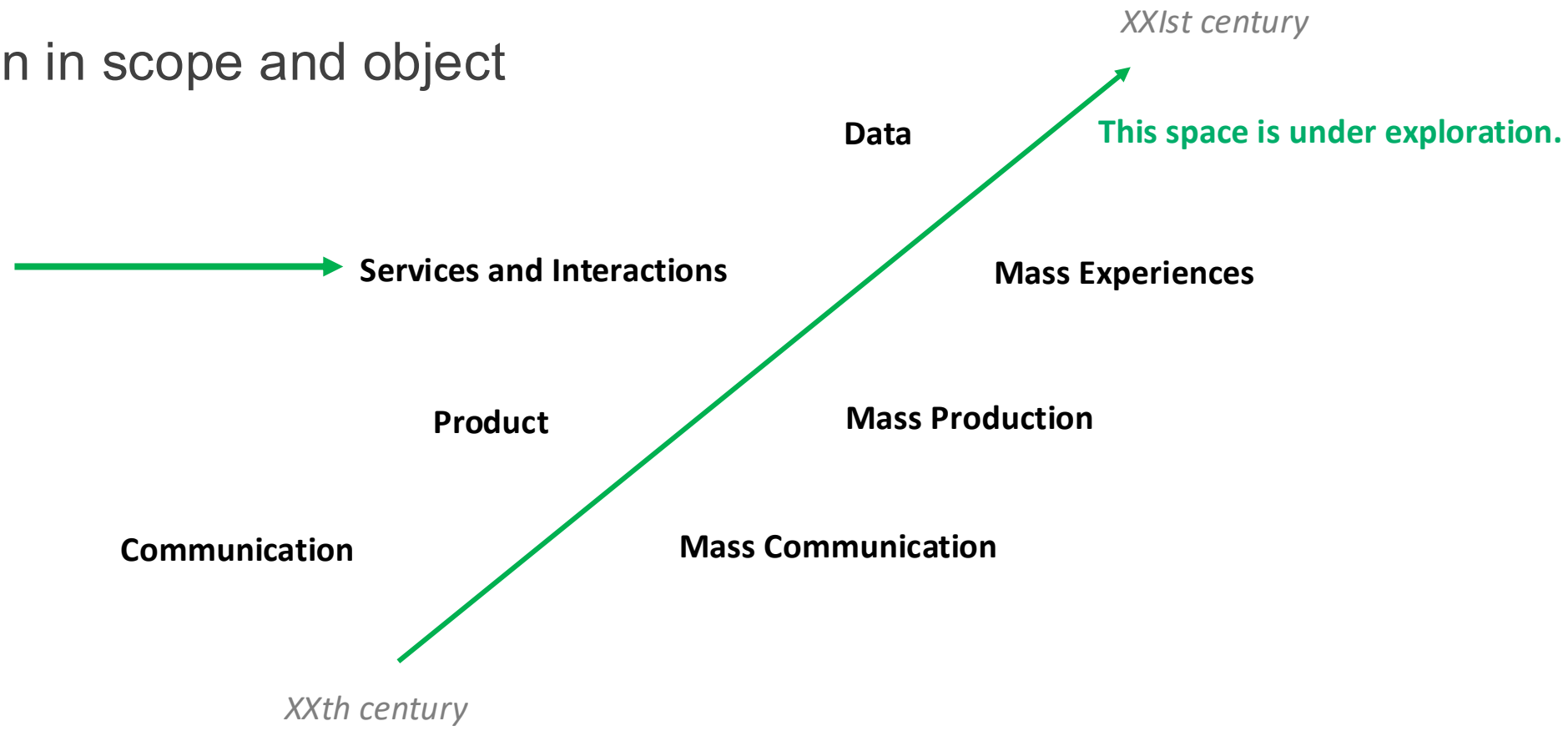
Design

Evolution in scope and object



Design

Evolution in scope and object



Design + Service

Emerged in the 90s, Service Design was initially focused on differentiating itself from industrial design (Holmlid & Evenson, 2008).

For a long time, Service Design developed as opposed to the Goods-Dominant Logic and under the umbrella of the newly proposed Service-Dominant Logic.

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2.1 DEFINING SERVICE DESIGN

Service design can help solve some important challenges faced by organizations. You hold a book in your hand which tells you how to do service design. But what does that term mean? It is something about customer experience, and innovation, and collaboration – but does it include everything related to those concepts, or is it just part of those worlds? Is every activity concerned with creating, planning, fixing, and shaping services part of "service design"? Do service designers even agree on what they do? Some people like to start with a definition, so in mid-2016 we asked 150 service designers to share and vote on their favorites. Here are the most popular:

- "Service design helps to innovate (create new) or improve (existing) services to make them more useful, usable, desirable for clients and efficient as well as effective for organizations. It is a new holistic, multi-disciplinary, integrative field." – *Stefan Moritz*[01]

- "Service design is the application of established design process and skills to the development of services. It is a creative and practical way to improve existing services and innovate new ones." – *live|work*[02]

- "Service design is all about making the service you deliver useful, usable, efficient, effective and desirable." – *UK Design Council*[03]

- "Service design choreographs processes, technologies and interactions within complex systems in order to co-create value for relevant stakeholders." – *Birgit Mager*[04]

- "[Service design is] design for experiences that happen over time and across different touchpoints." – *Simon Clatworthy, quoting* servicedesign.org[05]

- "When you have 2 coffee shops right next to each other, selling the exact same coffee at the exact same price, service design is what makes you walk into the one and not the other, come back often and tell your friends about it." – *31Volts*[06]

01 Moritz, S. (2005). *Service Design: Practical Access to an Evolving Field*. Köln.

02 live|work (2010). "Service Design." Retrieved 10 August 2010 from http://www.lbwork.co.uk.

03 UK Design Council (2010). "What Is Service Design?" Retrieved 10 August 2010 from http://www.designcouncil.org.uk/about-design/types-of-design/service-design/what-is-service-design/.

04 See for example "Meet Birgit Mager, President of the Service Design Network." Retrieved 3 August 2017 from https://www.service-design-network.org.

05 servicedesign.org is no longer accessible, but see Clatworthy, S. (2011). "Service Innovation through Touchpoints: Development of an Innovation Toolkit for the First Stages of New Service Development. International Journal of Design, 5(2), 15–28.

06 See 31Volts, "Service Design" (original quote from 2008 extended in 2014). Retrieved 3 August 2017 from http://www.31volts.com/en/service-design/.

Service Design

There are several terms used in the literature that address ideas related to how service organizations design new service offerings from either the customer's viewpoint or the delivery organization's viewpoint.

The most recent is **New Service Development** (NSD) or "**the overall process of developing new service offerings**" (Johnson et al., 2000).

The notion of **Service Design** has been more narrowly defined by Gummesson (1991), as "**the concretization of the service concept in drawings, flowcharts . . .** ".

Service Design

***Service design (SD) refers to the process of transforming the materials of service* (Blomkvist, Clatworthy, and Holmlid 2016).**

Early service management literature identified the focus of service design on **tangible evidence** (Shostack 1982) as well as on depicting the influence of the physical environment, or servicescape, on people engaged in service interactions (Bitner 1992).

In the 90s, SD built on User-Centred Design and Human-centred Design to depict **services physical evidences and interactions**.

Service Design

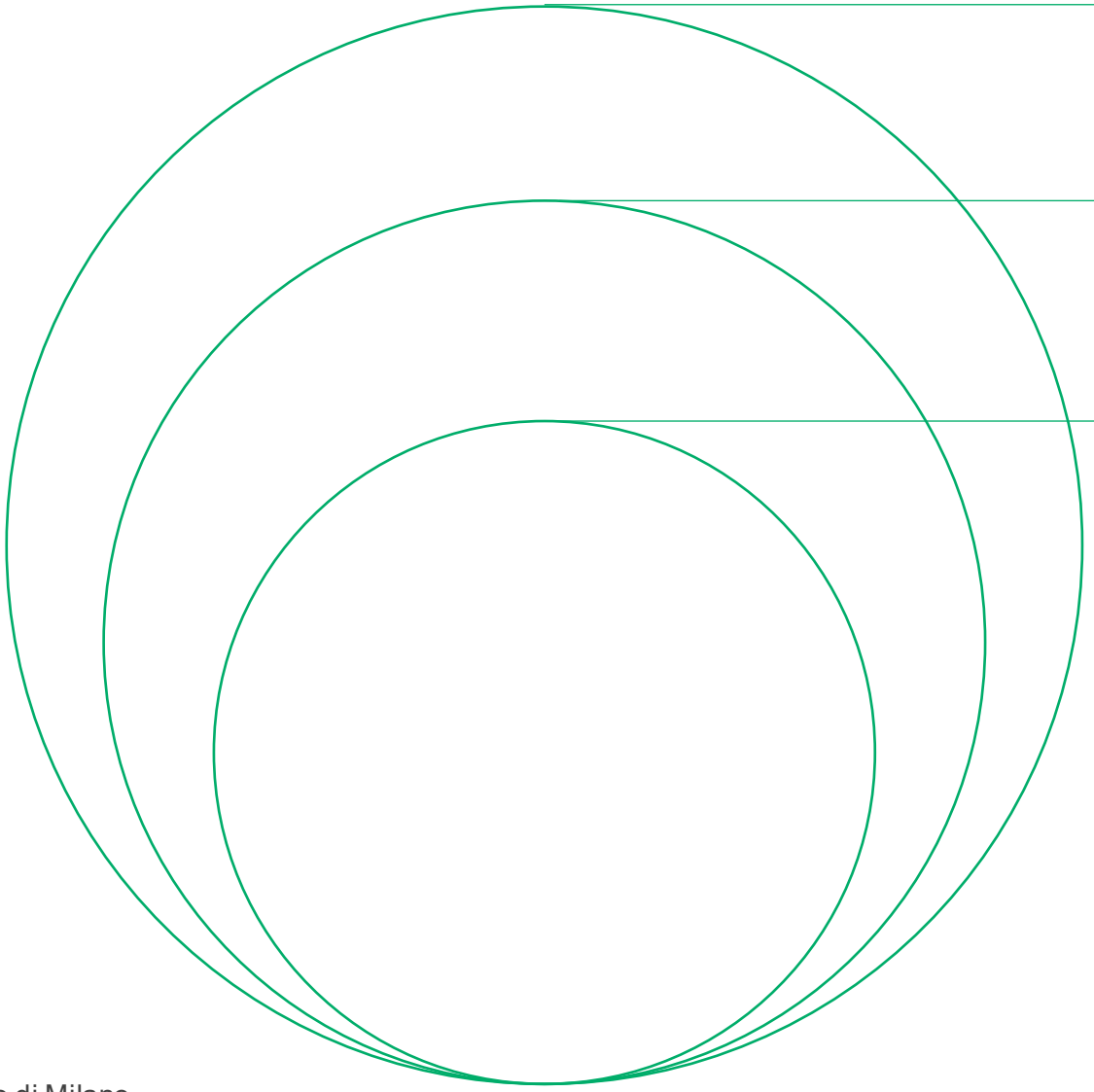
In the first decade of 2000, a new approach was proposed.

Design for Service (DFS) is a theorization in which the product-service distinction is no longer relevant, and the focus lies on value creation (Wetter-Edman et al., 2014).

Service Design

Recognising the shifts in S-D logic we explored earlier, recent service design literature shifted attention to **studying and depicting complex systems and service ecosystems** (Vink et al. 2021; van der Bijl-Brouwer 2022), focusing on the orchestration of the interconnections shaped by humans, digital systems, and organisations, propelled by the dynamics arising from the interplay between their respective behaviours.

Service Design



Designing institutional arrangements & infrastructures (macro-level)

Designing service systems (meso-level)

Designing interactions (micro-level)

Service Design

Many of these service ecosystems embed AI to a certain extent, thereby exerting a profound influence on the attributes of the entities that service design molds, as well as the interactions that service designers define in service delivery.

Consequently, scholars (Stoimenova and Price 2020; Verganti, Vendraminelli, and Iansiti 2020) have begun proposing that designers of these systems face not only ill-defined problems (Buchanan 1992) but also ill-defined systems (Mortati 2022), requiring to **understand AI as both a co-designer (or co-pilot) and another stakeholder in the system** (Giaccardi and Redström 2020).

Service Design

Emerging notions that you can explore:

Transition Design

Terry Irwin (2015) Transition Design: A Proposal for a New Area of Design Practice, Study, and Research, Design and Culture, 7:2, 229-246, doi: 10.1080/17547075.2015.1051829

Design and AI

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Fifth Order of Design

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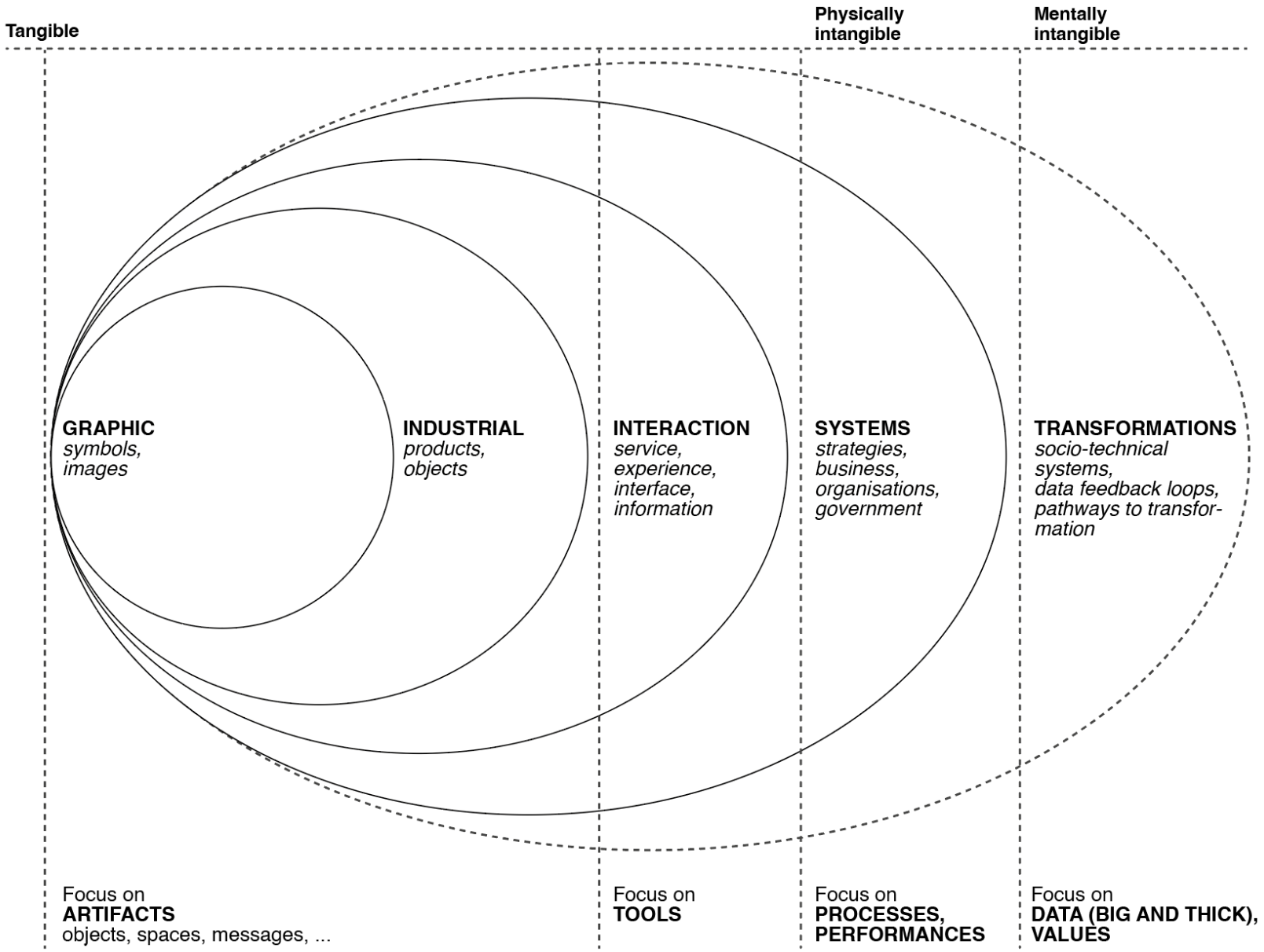
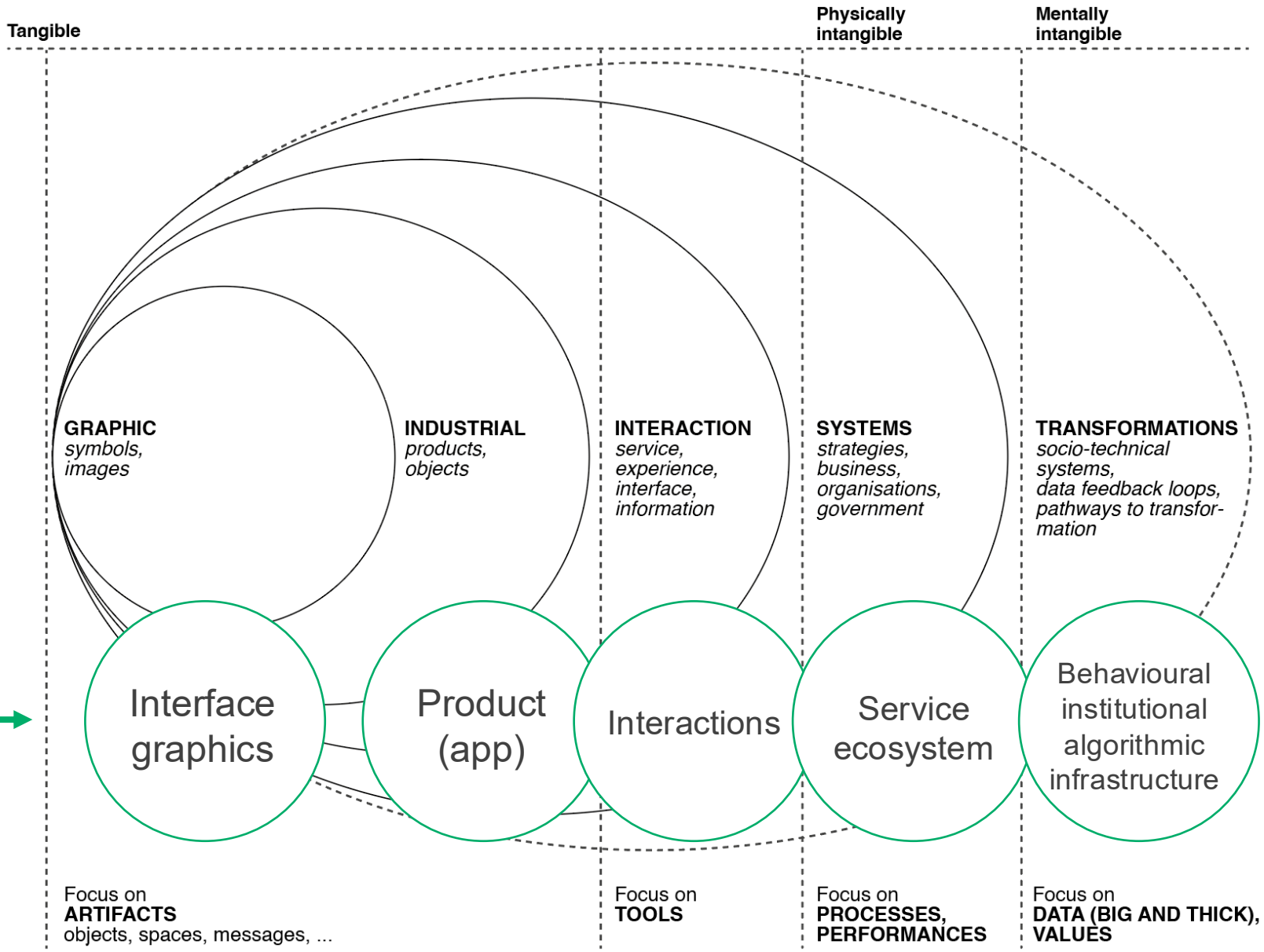
Design



(Source: Mortati, 2022)

Design

Spotify

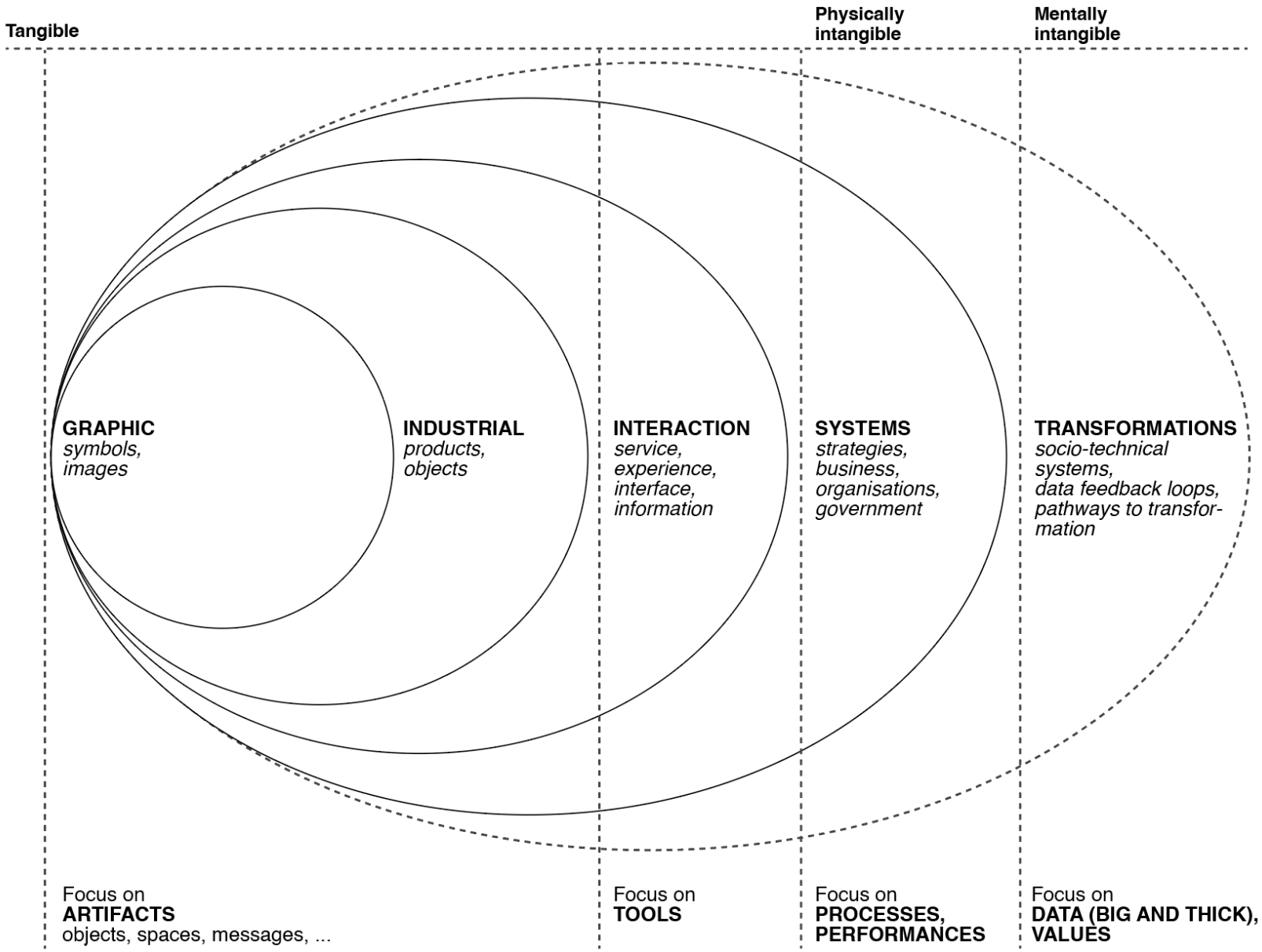


Design

In the Fifth Order of Design, we are no longer designing artefacts or evidences. **We are designing conditions.**

Conditions for:

- Participation
- Exclusion
- Visibility
- Agency
- Automation
- Governance



If design now operates at the level of systems and infrastructures… what kind of responsibility does that give designers?

If design now operates at the level of systems and infrastructures… what kind of responsibility does that give designers?

The fifth order is not just structural, it is ethical and political as designers increasingly help shape systems of behaviour.

Suggested readings

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THANK YOU